Reading the Captions

Each figure shows snapshots of the embedding during training. The caption describes which variant of the pipeline produced it, using the following terms.

HIGH DIMENSIONAL / LOW DIMENSIONAL Where the drift field is computed. In the *high dimensional* variant, the drift ω_i is computed in the ambient (original) space, the endpoints $x_i + \omega_i$ are added as virtual points, and the $2n$ real and virtual points are embedded in low dimension. The differences between virtual and real points give the drift b_i, which is then fed to the Randers force-directed layout (locate + apply pipeline). In the *low dimensional* variant there is no locate step: the drift $b_i = \frac{1}{k}\sum_j N_{ij}\, e_{ij}(Y)$ is computed directly in the low-dimensional embedding Y from the asymmetry matrix N.

FIXED / LIVE How the drift is used during training. With *fixed* drift, b_i is computed once before training and kept constant. With *live* drift, b_i is recomputed at every epoch from the current embedding Y.

PERTURBED / NOT PERTURBED Whether the IsUMap perturbation is applied to the k-NN distances. When perturbed, each distance is rescaled as $d_i = (d - \rho_i)/\sigma_i$, where ρ_i is the nearest-neighbour distance and σ_i the k-th neighbour distance of node i. This makes the distance matrix asymmetric. When not perturbed, the plain k-NN distances are used.

NOT STRONGLY CONNECTED The connectivity check on the directed k-NN graph uses weak connectivity (as in the earlier version) instead of strong connectivity.

FR GRAVITY The attractive/repulsive forces follow the Fruchterman–Reingold with gravity model (Bannister et al.) instead of the default UMAP force model with fitted curve parameters a, b.

N CHANGED The asymmetry matrix is computed as the bounded ratio $N_{ij} = (D_{ij} - D_{ji})/(D_{ij} + D_{ji})$ instead of the squared form $(D_{ij}^2 - D_{ji}^2)/(D_{ij} + D_{ji})$, which is equivalent to $D_{ij} - D_{ji}$ and is unbounded and scale-dependent.

Unless stated otherwise in the caption, the remaining settings are the defaults. Points are coloured by digit label for MNIST and by the height coordinate z (tail to head, viridis colormap) for the mammoth.

1 MNIST DATASET

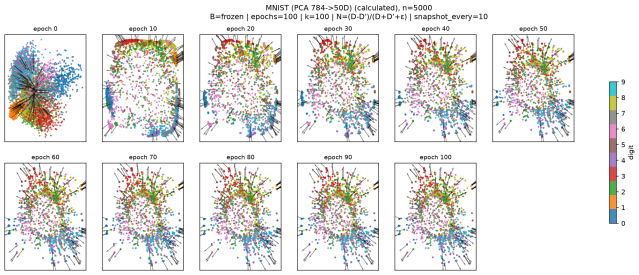


Figure 1: HIGH DIMENSIONAL, PERTURBED,FIXED

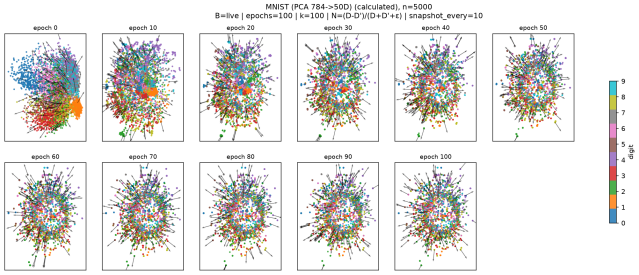


Figure 2: HIGH DIMENSIONAL, PERTURBED,LIVE



Figure 3: LOW DIMENSIONAL, NOT PERTURBED, FIXED

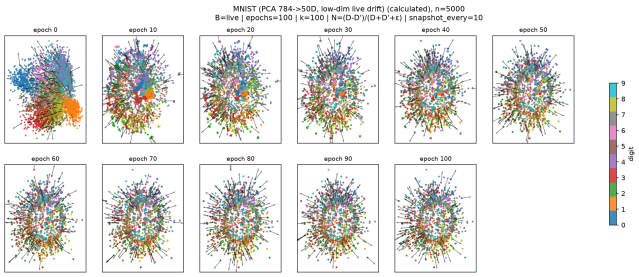


Figure 4: LOW DIMENSIONAL, NOT PERTURBED, LIVE

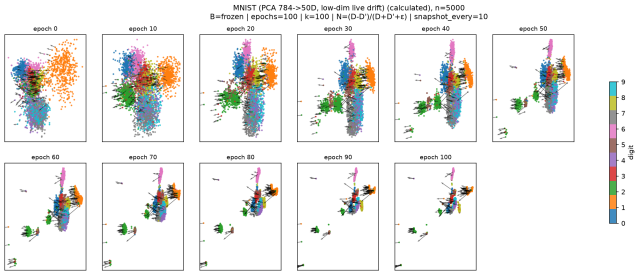


Figure 5: LOW DIMENSIONAL, PERTURBED, FIXED

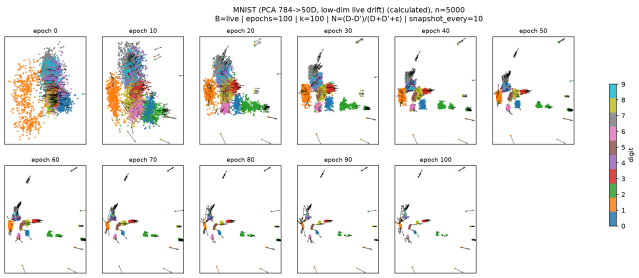


Figure 6: LOW DIMENSIONAL, PERTURBED, LIVE

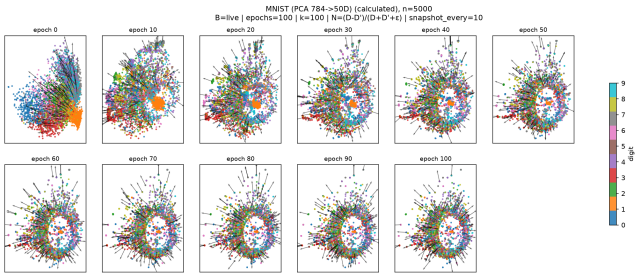


Figure 7: HIGH DIMENSIONAL, NOT PERTURBED,LIVE

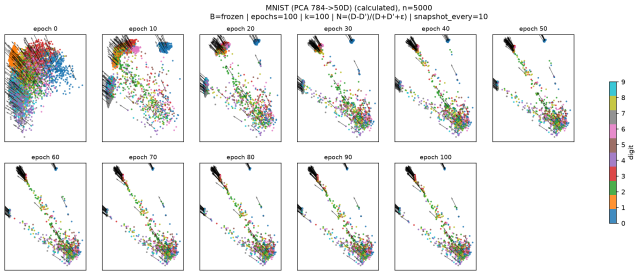


Figure 8: HIGH DIMENSIONAL, NOT PERTURBED,FIXED

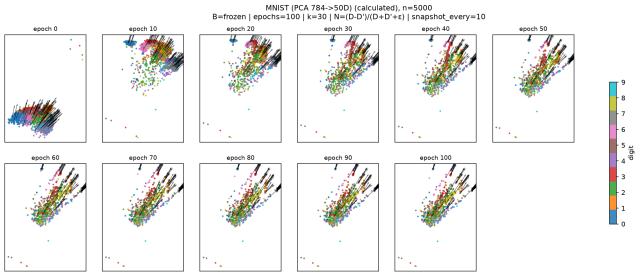


Figure 9: HIGH DIMENSIONAL, NOT PERTURBED,FIXED, not strongly connected

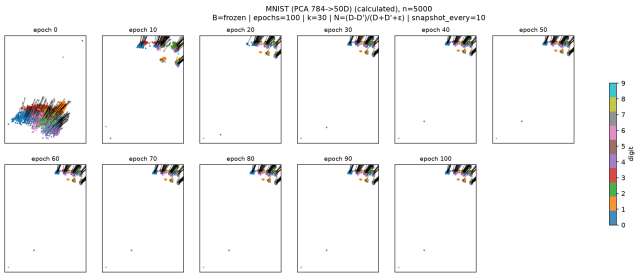


Figure 10: HIGH DIMENSIONAL, NOT PERTURBED,FIXED, FR Gravity

2 MAMMOTH DATASET

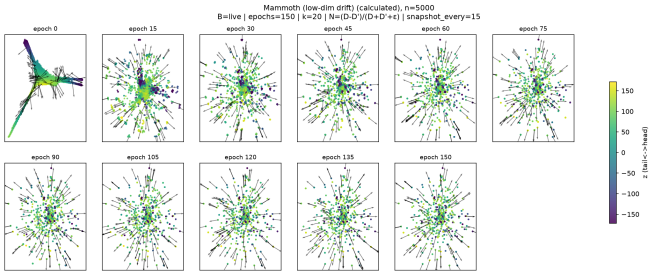


Figure 11: LOW DIMENSION, NOT PERTURBED, LIVE

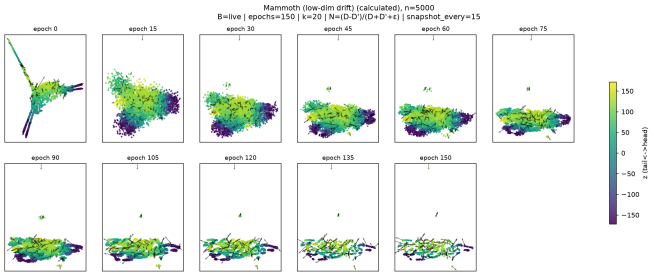


Figure 12: LOW DIMENSION, PERTURBED, LIVE

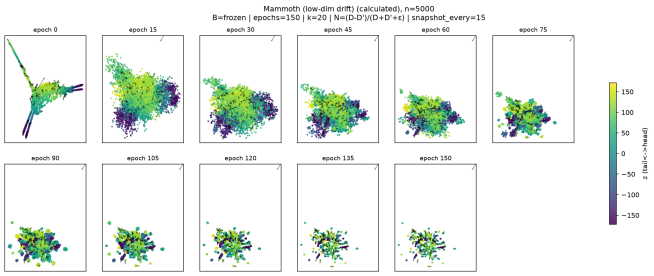


Figure 13: LOW DIMENSION, PERTURBED, FIXED

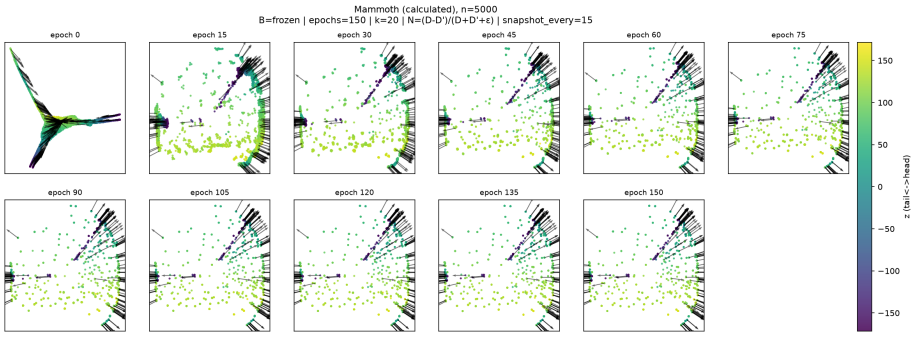


Figure 14: HIGH DIMENSIONAL, PERTURBED, FIXED

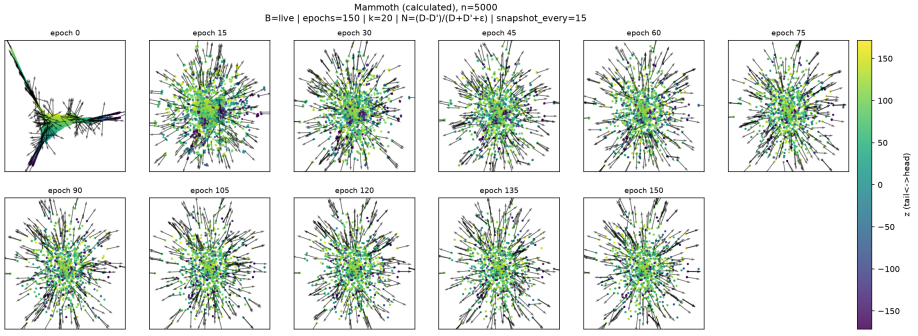


Figure 15: HIGH DIMENSION, PERTURBED, LIVE

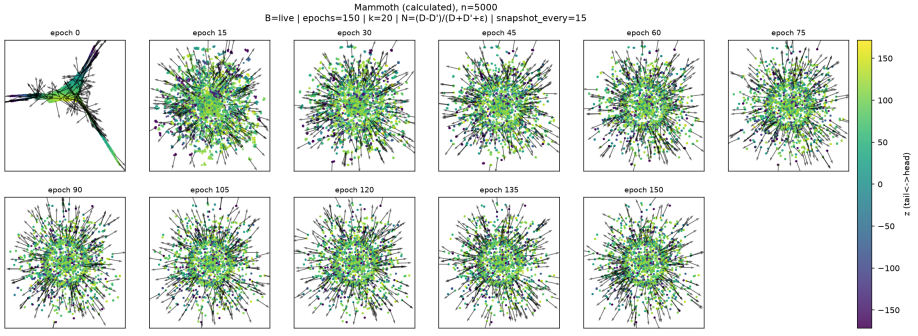


Figure 16: HIGH DIMENSIONAL, NOT PERTURBED, LIVE

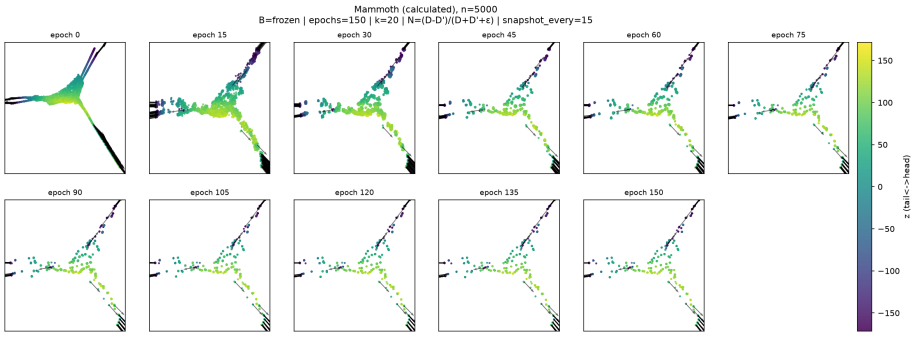


Figure 17: HIGH DIMENSIONAL, NOT PERTURBED,FIXED

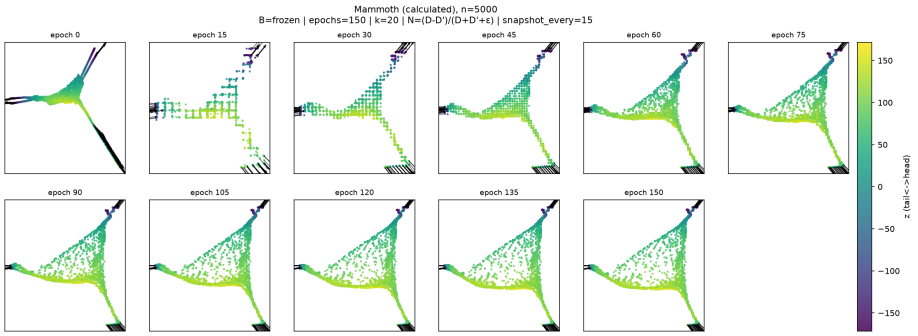


Figure 18: HIGH DIMENSIONAL, NOT PERTURBED,FIXED, FR GRAVITY

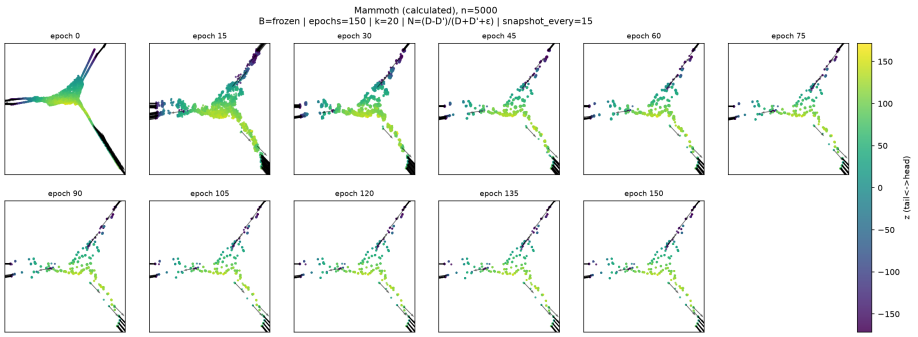


Figure 19: HIGH DIMENSIONAL, NOT PERTURBED,FIXED, NOT STRONGLY CONNECTED

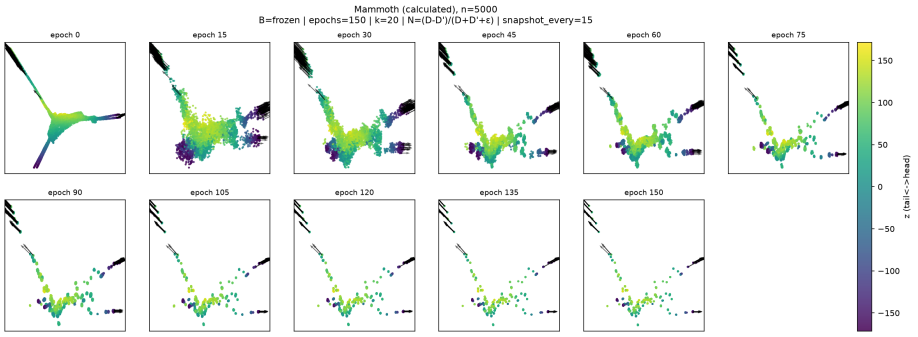


Figure 20: HIGH DIMENSIONAL, NOT PERTURBED,FIXED, N CHANGED